

PH 502: Optics/Photonics

Course Code	PH 502 Course Name: Optics/Photonics
Course Name	Optics/Photonics
L-T-P-C	3-0-0-3
Prerequisites	Foundations of Electrodynamics & Quantum mechanics
Intended for	B.Tech
Distribution	elective
Approval	2nd Adhoc Senate (May 2012), in the Minor in Applied Physics curriculum

Course Contents

- **Electromagnetic Optics**
 - Electromagnetic theory of light, electromagnetic waves in vacuum & dielectric media, absorption and dispersion, pulse propagation in dispersive media, Metamaterials [6 lectures]
- **Polarization Optics**
 - Polarization of light, reflection and refraction, optics of anisotropic media, Optics of liquid crystals, polarization devices. [5 lectures]
- **Guided wave Optics**
 - Electromagnetic waves in dielectric layered media, photonic crystals, waveguides, resonators, plasmonics. [5 lectures]
- **Fiber Optics**
 - Electromagnetic waves in fiber, Attenuation and dispersion, photonic crystal fibers. [5 lectures]
- **Semiconductor Optics**
 - Quantization of electromagnetic field, quantum states of light, photon statistics, interaction of photons with charge carriers, light emitting diodes, laser diodes, microcavity lasers. [6 lectures]
- **Detection of light**
 - Theory of photo detection, photodetectors, photodiodes, avalanche photodiodes, noise in photodetectors. [5 lectures]
- **Acousto and Electro Optics**
 - Interaction of light and sound, acousto-optic devices, Principles of electro optics, electro optics of anisotropic media, electro optics of liquid crystals. [5 lectures]
- **Optical fiber communication**
 - Fiber Optic components, optical fiber communication system, modulation and multiplexing, fiber optic networks. [5 lectures]

Text Books

1. A.K. Ghatak, K. Thyagarajan, **Optical Electronics**, Cambridge University Press

References

1. Max Born, Emil Wolf, **Principles of Optics**, Cambridge University Press.
2. Saleh & Teich, **Fundamentals of Photonics**, Wiley-Inter science.